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MILITARY SPECIFICATION

CUTTER, POWDER ACTUATED, M8 ASSEMBLY

1. SCOPE

1.1 This specification is the applicable detail specification for Cutter, Powder Actuated, M8, Assembly, loaded with Cartridge, Cutter, M122 for use in conjunction with other propellant actuated devices in aircrew emergency escape systems.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, or request for proposals form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-2550 — Ammunition and Special Weapons, General Specification for.

MIL-W-8777 — Wire, Electrical, Copper, 600 Volts, 150° Centigrade.

MIL-W-12349 — Wire, Hook-Up, Monochlorotrifluoroethylene Insulated.

MIL-I-45208 — Inspection System Requirements.

MIL-C-17/30 — Cable, Radio Frequency, Coaxial, RG-62 A/U.

MIL-P-19834 — Plates, Identification, Metal Foil, Adhesive Backed.

STANDARDS

MILITARY

MIL-STD-105 — Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-109 — Quality Assurance Terms and Definitions.

MS28741 — Hose Assembly—Detachable End Fitting, Medium Pressure.

DRAWINGS

MUNITIONS COMMAND

F10520566 — Cutter, Powder, Actuated, M8, Assembly.

FSC 1375

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C8835919 — Container, Ammunition, Fiber M346 for Cutter, Cartridge Actuated, M8 with Cartridge M122.

C8835920 — Box, Packing, Ammunition for Cutter, Cartridge Actuated, M8 with Cartridge M122 in Fiber Container M436.

IEL-10520566 — Index of Inspection Equipment List for Cutter, Powder Actuated, M8, Assembly.

(Copies of specifications, standards and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General. Material, parts and assemblies shall comply with the requirements specified on the applicable drawing, referenced specifications and the following:

3.2 Torque.

3.2.1 *Breakaway torque.* Torque requirements shall be as specified in table I. No movement or deformation shall occur upon application of the breakaway torque.

TABLE I

Component	A Run on, min. (inch-pounds)	B Seating or bottoming including A (inch-pounds)	C Breakaway, min. inch-pounds)
Body and head after insertion of cartridge	40	600 ± 30	400

3.2.2 *Rotating torque.* Upon completion of assembly the swivel shall rotate 360 degrees in either direction when a torque of 75 inch pounds plus or minus 25 inch pounds is applied between the swivel and body.

3.3 Residual magnetism. The cutter shall not deflect the indicator of the compass more than 3 degrees in either direction when passed in front of the compass at a distance of 6 inches.

3.4 Functional.

3.4.1 The cutter shall not fail to fire.

3.4.2 *Load.* The cutter fired under load shall completely sever any one of the 2 groups of electrical conductors as detailed below (see 6.4).

3.4.2.1 *Cables.* A .060 inch thick vinyl sheathed cable assembly consisting of 9 RG-62 A/U Coax Cables (MIL-C-17/30).

3.4.2.2 *Wires.* A .060 inch thick vinyl sheathed bundle composed of 27 strands of 22 gage wires (MIL-W-8777), 6 strands of 18 gage wires (MIL-W-8777) and 8 strands of 20 gage wires (MIL-W-12349).

3.4.3 *Locked shut.* The cutter fired in a locked shut condition shall retain the generated gases and show no evidence of mechanical failure (see 6.1 and 6.3).

3.4.4 *No load.* The cutter fired with no load shall show no mechanical failure (see 6.1, 6.5 and 6.6).

3.4.5 *Temperature.* The cutter shall be capable of complying with all the listed functional requirements throughout the temperature range of minus 65 to plus 200 degrees F.

3.4.6 *Initiation.* The cutter shall be initiated with an M3A1 Initiator connected to the cutter with 15 feet of aircraft hose, size 4 Standard MS28741.

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3.4.7 *Ignition delay.* The cutter when fired shall have an ignition delay of .025 seconds, maximum (see 6.2).

3.5 **Workmanship.** The requirements for workmanship are as specified on the applicable drawings, referenced specifications and the following:

3.5.1 *Bur.* No part shall have a bur which will interfere with the assembly or function of the item or which might be injurious to personnel handling the item.

3.5.2 *Foreign matter.* No part or assembly shall contain chips, dirt, grease, rust, corrosion or other foreign matter.

3.5.3 *Label, edge adhesion.* The labels shall withstand two or more traverses of a knife at any point without damage. There shall be no indications of poor adhesion; such as nicking, tearing, or peeling of the labels in whole or in part from the thruster.

4. QUALITY ASSURANCE PROVISIONS

4.1 **Responsibility for inspection.** Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 *Quality assurance terms and definitions.* Reference is made to Standard MIL–STD–109 to define quality assurance terms used.

4.1.2 *Inspection.* Inspection shall be in accordance with Specifications MIL–A–2550 and MIL–I–45208, except as specified herein.

4.2 **Inspection provisions.**

4.2.1 *Lot.*

4.2.1.1 *Submission of product.* The product shall be submitted in accordance with Standard MIL–STD–105.

4.2.1.2 *Size of lot.* A lot shall consist of an identifiable group of assembled units containing metal parts assemblies produced by a homogeneous process and loaded with a single lot of cartridges.

4.2.2 *Examination.* One hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed on a defect basis in accordance with the classification of defects contained herein, using the sampling plan and associated criteria in table II. The total number of allowable defects per class for either the major or minor classification shall not exceed the acceptance number of table II. All non-conforming material shall be rejected. Four power (4X) magnification may be used in performing visual examination for such defects as burr and foreign matter.

TABLE II

Lot size	Sample size	Major		Minor	
		Accept	Reject	Accept	Reject
1 – 300	100% Examination				
301 – 800	50	0	1	1	2
801 –1800	90	1	2	2	3

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4.2.2.1 *Cartridge, Cutter, M122, Dwg. D10520599, covering a detail of Dwg. F10520566.*

Defect	Method of Inspection
Critical:	
1. Mixed cartridges	Visual
2. Metal defective	Visual
3. Marking missing, incorrect or illegible	Visual
Major:	
101. Foreign matter	Visual
Minor:	
201. Case scratched or dented	Visual

4.2.2.2 *Blade, Subassembly, Dwg. F1052056 6, less the following components Shear Pin 8593876, Head 10520573, Firing Pin 10520574, Cartridge 10520599, BCTX2.15–588861, MS29513–12 and MS29513–1 28.*

Critical:	
1. Presence of shear pin in blade	Manual/Visual
Major:	
101. Protective finish inadequate or defective	Visual
102. Burr	Visual
103. Foreign matter	Visual
Minor:	
None defined	

4.2.2.3 *Head, Subassembly, Dwg. F10520566, less all components except Shear Pin 8593876, Head 10520873, Firing Pin 10520574, BCTX2.15–588861 and MS29513–12.*

Critical:	
1. Shear pin not engaged in firing pin	Gage
Major:	
101. Any part missing	Visual
102. Thread damaged	Visual
103. Protective finish inadequate or defective	Visual
104. Burr	Visual
105. Foreign matter	Visual

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Defect	*Method of Inspection*
Minor:	
201. Set screw loose	Manual
202. Lubricant on "O" ring, missing or excessive	Visual

4.2.2.4 *Body and Cartridge, Subassembly, Dwg. F10520566, less Head, Subassembly.*

Critical:	
1. Mixed cartridges	Visual
2. Any part missing, except as otherwise specified	Visual
3. Improper assembly	Visual
Major:	
101. Thread damaged	Visual
102. Bur	Visual
103. Foreign matter	Visual
Minor:	
201. Pellet missing	Visual

4.2.2.5 *Assembly. Dwg. F10520566.*

Critical:	
None defined	
Major:	
101. Length from face of cap to face at gas port	Gage
102. Sealing compound on thread, at head, missing	Visual
103. Mounting nut missing	Visual
104. Marking missing, incorrect or illegible	Visual
105. Thread damaged	Visual
Minor:	
201. Set screw above flush	Visual
202. Protective finish inadequate or defective	Visual
203. Label missing	Visual
204. Shipping plug missing	Visual
205. Car seal missing or embossed	Visual

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4.2.2.6 *Fiber Container, Dwg. C8835919*

Defect	*Method of inspection*
Critical:	
None defined	
Major:	
None defined	
Minor:	
201. Gap between body and cover, max.	Manual
202. Contents loose	Manual
203. Metal ends loose or distorted	Manual
204. Cut or damaged through all impregnated layers	Visual
205. Marking missing, incorrect or illegible	Visual
206. Tape incomplete or badly wrinkled	Visual
207. Length of tear tab, min.	Visual
208. Color of tape incorrect	Visual

4.2.2.7 *Box, Packing, Dwg. C8835920.*

Critical:	
None defined	
Major:	
None defined	
Minor:	
201. Contents loose	Manual
202. Hardware missing, loose, broken or improperly engaged	Manual/Visual
203. Strapping missing, loose, broken or improperly assembled	Manual/Visual
204. Contents of box exposed	Visual
205. Box damaged	Visual
206. Marking misleading or unidentifiable	Visual
207. Number of units incorrect	Visual
208. Car seal missing, unsealed or improperly positioned	Visual

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4.2.3 *Inspection testing.* The following tests shall be performed in accordance with 4.3 except as provided in Specification MIL-I-45208.

4.2.3.1 *Torque.*

4.2.3.1.1 *Breakaway torque.* Each body and head assembly shall be subjected to a breakaway torque application. Failure of any assembly to comply with the requirements of 3.2.1 and table I shall be cause for rejection of the assembly.

4.2.3.1.2 *Rotating torque.* Each body and swivel assembly shall be subjected to a rotating torque application. Failure of any assembly to comply with the requirements of 3.2.2 shall be cause for rejection of the assembly.

4.2.3.2 *Residual magnetism.* Each cutter assembly shall be subjected to a test for excessive residual magnetism. Failure of any assembly to comply with the requirement of 3.3 shall be cause for rejection of the assembly.

4.2.3.3 *Label, edge adhesion.* Ten assemblies from each lot shall be tested for adhesion of the label edges. Failure of any unit to comply with the requirements of 3.5.3 shall be cause for rejection of the lot of thrusters.

4.2.3.4 *Functional tests.* The following tests shall be performed by the Government.

4.2.3.4.1 *Ballistics.* Eighteen units shall be tested to determine compliance with the requirements of 3.4.1 thru 3.4.7. The units shall be temperature conditioned as provided in table III and 4.3.4.1.1. Failure of any unit to comply with the requirements shall be cause for rejection of the lot.

TABLE III

Sample size	Method	Temperature
7	Load	−65° to −70°F.
7	Load	200° to 205°F.
1	No Load	−65° to −70°F.
1	No Load	200° to 205°F.
1	Locked Shut	−65° to −70°F.
1	Locked Shut	200° to 205°F.

4.2.4 *Inspection equipment.* The examination and tests shall be made using equipment listed on IEL-10520566, except as provided in Specification MIL-I-45208.

4.3 Test methods and procedures.

4.3.1 *Torque.* Torque wrenches, set at the torque requirements specified in table I and 3.2.2, shall be used in conjunction with a fixture to determine the adequacy of the joint. The cutter body shall be secured in a holding device. The torque shall be applied to the head and cap, as applicable.

4.3.2 *Residual magnetism.* The cutter shall be tested for excessive residual magnetism in an area free of local magnetic effects by placing the cutter 6 inches from and at the same height as the compass, in a north-south horizontal position with the compass set in an east-west heading. The cutter shall be moved slowly past the compass in a longitudinal direction for its entire length, rotated 90 degrees in the horizontal position about its longitudinal axis and moved slowly back past the compass. This procedure shall be followed for each 90 degree rotation.

4.3.3 *Label, edge adhesion.* The label, edge adhesion test shall be in accordance with Specification MIL-P-19834.

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4.3.4 *Functional.*

4.3.4.1 *Ballistics.* The test samples shall be rigidly supported so that the entire thrust is borne by the mounting nuts. The sample shall be tested within 5 minutes after removal from the temperature conditioning chamber. When this limit is exceeded the sample units shall be reconditioned at the original temperatures.

4.3.4.1.1 *Temperature conditioning.* A simulated cutter, loaded with a dummy M122 Cartridge, containing a temperature sensitive element, shall be stored in a temperature conditioning chamber at the same time as the samples. The temperature conditioning time shall be a minimum of one hour longer than the time necessary for the temperature of the dummy cartridge to reach the test temperature. The total conditioning time including reconditioning at plus 200 degrees F. shall not exceed 18 hours.

4.3.5 *Ignition delay.* The cutter ignition delay shall be recorded by instrumentation at the inlet port.

5. PREPARATION FOR DELIVERY

5.1 **Packing and marking.** Level A—(World Wide Shipment) — Packing and marking shall be in accordance with Drawings C8835919 and C8835920.

5.2 **Data cards.** Data card information shall be as specified in Specification MIL–A–2550.

6. NOTES

6.1 A mechanical failure is defined as any deformation or breakage of a part, the occurrence of which is other than a design function.

6.2 Ignition delay is defined as the time lapse between energizing of the primer and the first continuous pressure rise.

6.3 Locked shut is defined as a condition which prevents the blade from moving upon firing.

6.4 **Load.** The expected performance of the cutter under conditions simulating those existing in aircraft is obtained when the cutter satisfactorily meets conditions of the load firing.

6.5 No load is defined as a condition in which there are no wires, cables, or any other obstacle between the blade and anvil.

6.6 Permanent yielding of swivel shall not be cause for rejection when cutters are tested with no load opposing the blade.

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